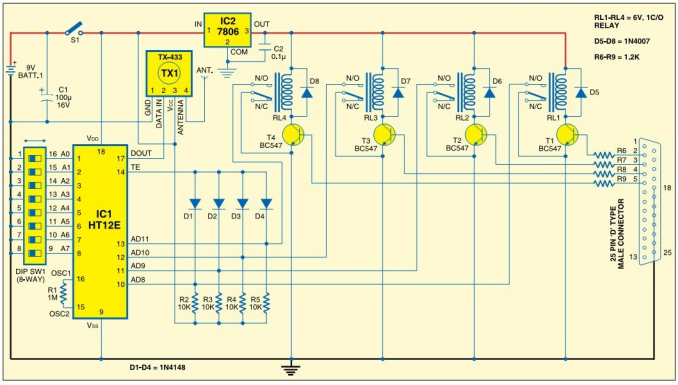


Fig. 4: Transmitter circuit for wireless stepper motor control

encoder HT12E. The values of timing resistors connected between OSC1 and OSC2 pins of HT12E and HT12D, for the given supply voltages, can be found out from the graphs given in the datasheets of the respective chips (included in this month's EFY-CD). The resistance values used in the circuits are chosen here for approximately 3kHz frequency of the encoder (HT12E) at Vcc of 9V and 150 kHz of the decoder (HT12D) at Vcc of 5V.

The HT12D receives the data from the HT12E on its D_{IN} pin serially. If the address part of the data received matches the levels on A0 through A7 pins four times in succession, the valid transmission pin (V_T) is taken high. The data on pins AD8 through AD11 of the HT12E appear on pins D8 through D11 of the HT12D. Thus the device acts as a receiver of 4-bit data (16 possible codes) with 8-bit addressing (256 possible channels).

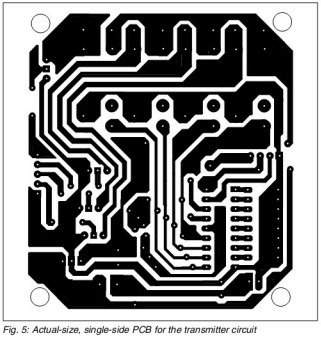


Fig. 5: Actual-size, single-side PCB for the transmitter circuit

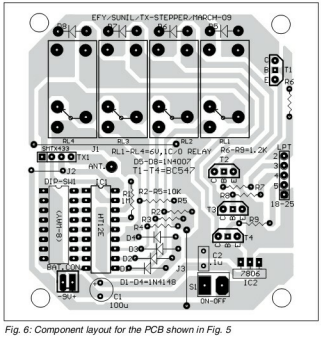


Fig. 6: Component layout for the PCB shown in Fig. 5